Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Software engineer with 8+ years shipping scalable production systems, specializing in generative AI agents and agentic orchestration on LangChain and LangGraph. Built **AgentProbe** - a LangSmith-equivalent open-source evaluation and observability framework: 5-metric eval engine (correctness, grounding, latency, token efficiency, hallucination detection), CI-integrated regression gate, and multi-provider LLM routing - used to validate production AI agent quality across 1,240+ test cases. Built **Leasel**: an 8-agent LangGraph multi-step workflow with RAG-based retrieval, conditional orchestration, failure handling via HITL escalation, and grounding verification - 93.6% precision on 1,240 compliance documents. At Entrata, owned Python pipelines and production microservices at 100K+ MAU scale across 120+ Agile sprints. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). [AgentProbe: github.com/venkatalakshmiparimala/AgentProbe]

CORE COMPETENCIES

LangChain · LangGraph Production Agents | LangSmith-Equivalent Evaluation & Observability | Multi-Step Agentic Orchestration | Failure Handling · HITL · Grounding | Generative AI · RAG Pipelines | Python · JavaScript · FastAPI | Scalable Production Systems · CI/CD | Forward-Deployed Engineering · POC Delivery · Pre-sales Enablement | Technical Customer Delivery · Workshops

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern (Unpaid)

- Built an agentic LangGraph multi-step workflow (parse -> plan -> retrieve -> validate -> conditional routing) - a generative AI pipeline translating natural-language requests into structured flight plans with RAG-based grounding over FAA Part 107 regulations and deterministic failure handling for non-compliant missions.
- Engineered validation guardrails enforcing altitude limits, battery budgets, and geometric no-fly-zone checks - blocking non-compliant missions before export; every compliance decision made by deterministic logic, not the LLM.

George Mason University - College of Engineering

Aug 2025 - May 2026 · Part-time

Graduate Teaching Assistant

- Delivered technical demos, workshops, POC walkthroughs, and one-on-one debugging sessions to 600+ developers (students) across multiple sections covering Python programming, data structures, OOP, and database design - adapted explanations to varied skill levels in real time; equivalent experience to pre-sales technical enablement for developer audiences.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets across 190+ countries - $R^2 = 0.977$; applied Isolation Forest for automated data integrity checks across 3 research pipelines; designed Tableau and ggplot2 dashboards improving research comprehension 40%.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 - Aug 2024 · Remote

Senior Software Engineer

- **Shipped 20+ scalable production modules** (Python, PHP, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; led microservices redesign with Docker containerization and CI/CD pipelines - 50% throughput improvement, 30% deployment time reduction, 80% reduction in manual operational workload.
- **Owned a production data pipeline for 7 years** - federal source rows transformed into ~180,000 domain records powering eligibility, rent, and tax-credit calculations for 100K+ housing units; every release cycle included baseline validation, unit tests, reconciliation, and failure handling with rollback preparation. Zero compliance violations.
- **Built RabbitMQ-based async production system** for gross-rent updates across thousands of tenant records - designed retry logic and idempotency guarantees as failure handling for double-processing; owned through production incidents and annual changes.
- **Optimized 200+ SQL queries (30% performance gain)** through execution plan analysis, indexing, partitioning, and stored procedure refactoring across scalable multi-tenant production schemas.
- **Led AWS S3 document migration** - 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents across the full production migration.
- Mentored 5+ engineers; conducted 300+ peer code reviews; built debugging tools reducing debugging time 40% and keeping open bugs below 10 for 6+ consecutive months.

Entrata India (Db Xento Systems)

Jan 2017 - Sep 2018 · Pune

Associate / Trainee Software Engineer

- Built property management modules (PHP, Python, PostgreSQL) for compliance workflows; led AWS S3 cloud migration resolving 90% of legacy storage constraints; redesigned document storage database architecture for 180K+ records.

PROJECTS

AgentProbe: LangSmith-Equivalent Evaluation & Observability Framework

Open Source

2025 | Python, LangGraph, FastAPI, ChromaDB, CI/CD (GitHub Actions)

github.com/venkatalakshmiparimala/AgentProbe

Built a production-grade generative AI evaluation and observability framework: 15-endpoint FastAPI backend, 5-metric eval engine (correctness, grounding, latency, token efficiency, hallucination detection), CI-integrated regression gate blocking quality regressions before merge, and multi-provider LLM routing. Plug-and-play: any agent connects via API; LangSmith-compatible evaluation patterns with LLM-as-judge scoring. Validated Leasel across 1,240+ test cases: pass@1 81.7%, pass@3 88.3% via multi-trial sampling harness.

Leasel: 8-Agent LangGraph Multi-Step Agentic Workflow

Production Architecture

2026 | Python, LangGraph (8-agent), FastAPI, RAG, HITL, OCR, SQLite

github.com/venkatalakshmiparimala/Lease_complaince_rag

Built an 8-agent LangGraph multi-step orchestration pipeline (Triage -> Retrieval -> Configuration -> Historical Incident -> Structured Data -> Reasoning -> Validation -> Human Escalation) with RAG-based grounding and failure handling via HITL escalation when agent confidence is too low. Results: 93.6% precision, 87.9% F1 on 1,240 compliance documents; 100% citation correctness; 0% unsupported-claim rate. Core rule in code: LLM reasons, human or deterministic logic makes every final regulated call.

AI on FHIR: NLP-Powered Medical Records Query System

2024 | Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings, JavaScript

github.com/venkatalakshmiparimala/AI_on_FHIR

Reduced manual clinical retrieval time ~60% by building a Flask + React generative AI platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; enforced HIPAA-compliant RBAC. Containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering - Jawaharlal Nehru Technological University, Anantapur — 2017

SKILLS

Languages: Python, JavaScript, TypeScript, PHP, SQL, PL/SQL, R, Bash

Generative AI & Agents: LangChain, LangGraph, LangSmith-compatible Evaluation, Generative AI, Agentic Orchestration, Multi-Step Workflows, Failure Handling, HITL, RAG Pipelines, Grounding, LLM Evaluation, Observability, Multi-Agent Systems, Context Engineering, Prompt Engineering, ChromaDB, Vector Databases, Semantic Search, Embeddings, OpenAI API, Anthropic API, FastAPI, MCP

ML & Data: PyTorch, TensorFlow, XGBoost, Scikit-learn, PySpark, NumPy, Pandas, Tableau, Power BI, spaCy, BERT, Hugging Face, NLP

Infrastructure: AWS (S3, EC2, RDS, Glue), Docker (container orchestration), Kubernetes (fundamentals), CI/CD (GitHub Actions), Scalable Microservices, RabbitMQ, PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum